

ONLINE EVENT TICKET BOOKING SYSTEM

AI23431- WEBTECHNOLOGY AND MOBILE

APPLICATION

MINI-PROJECT REPORT

SUBMITTED BY

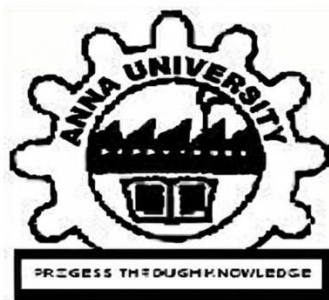
MITHUN CHAKRAVARTHY (241501158)

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE
AND DATA SCIENCE**

**RAJALAKSHMI ENGINEERING COLLEGE
(AUTONOMOUS),
CHENNAI-602 105**

APRIL 2025

RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project Thesis titled “**ONLINE EVENT TICKET BOOKING STSTEM**” is the Bonafide work of MITHUN CHAKRAVARTHY L (2116241501158) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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MITHUN CHAKRAVARTHY L
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DEPARTMENT VISION :

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION :

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Data Science Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and DS algorithms. To understand the standard practices and strategies in project development, using open- ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and DS professional.

COURSE OBJECTIVE :

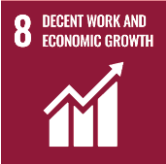
- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

COURSE OUTCOME :

On completion of course students will be able to,

- CO 2: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.
- CO 3: Design and implement dynamic web page with validation using javascript objects and by applying different event handling mechanism.
- CO 4: Design and implement simple webpage to learn JSP and Servlet.
- CO 5: Design and implement simple Application Design.
- CO 6: Identify various concepts of mobile programming that make it unique from programming for other

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
 SDG-4	Quality Education	Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.
 SDG-8	Decent Work and Economic Growth	Necessary for social stability, shared prosperity, and combating poverty.
 SDG-9	Industry, Innovation, and Infrastructure	Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.
 SDG-17	Partnerships for the Goals	Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.

ABSTRACT

The Online Event Ticket Booking System presents a digital solution for efficiently browsing, booking, and managing tickets for various events such as concerts, movies, sports, and cultural programs. With the increasing demand for convenient and real-time access to event information, the system is designed to replace traditional offline ticket booking methods with a secure and automated platform. The application enables users to explore events, check availability, and book tickets online, while administrators can manage event details and monitor bookings through a dedicated dashboard. By integrating role-based authentication, the system provides secure login access for both users and administrators, ensuring that booking data is protected and accessible only to authorized individuals.

The system includes features such as real-time ticket availability, secure online payment integration, and instant booking confirmation to enhance user experience. Users can select events based on category, date, or location, and reserve tickets with ease. The system ensures that duplicate bookings are avoided and seat availability is accurately maintained. Additionally, users receive automated notifications via email or SMS regarding booking confirmations, event updates, or cancellations, improving communication and transparency.

The architecture of the system is designed to be efficient and user-friendly, allowing smooth operation on standard systems without requiring complex infrastructure. By maintaining centralized data storage and providing analytics through dashboards, the system helps administrators track booking trends and manage event capacity effectively. Overall, the Online Event Ticket Booking System improves convenience, reduces manual workload, and enhances user satisfaction while ensuring reliable and secure ticket management.

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CHAPTER 1

INTRODUCTION

1.1 General

Event ticket booking plays an important role in the entertainment and service industry as it enables users to access and participate in various events such as concerts, movies, sports matches, and cultural programs. Traditionally, ticket booking is done through physical counters or manual processes, which can be time-consuming, inconvenient, and prone to errors. Manual systems also make it difficult to track ticket availability, manage large crowds, and generate booking reports efficiently.

The Online Event Ticket Booking System is designed to automate the process of browsing, booking, and managing event tickets through a digital platform. The system allows users to search for events, check seat availability, and book tickets online, while administrators can manage event details, schedules, and bookings through a secure dashboard. By maintaining booking data in a centralized database, the system ensures accuracy, easy access to information, and improved operational efficiency.

This system also reduces administrative workload and enhances user convenience by providing anytime, anywhere access to ticket booking services. With features such as secure login authentication, real-time availability tracking, and automated booking management, the system simplifies the overall ticket booking process. It also improves transparency by allowing users to view booking details and receive instant confirmations, making the experience more reliable and user-friendly.

1.2 Need for Student Attendance Management System

Many event organizers still rely on traditional ticket booking methods such as physical counters or manual reservations, which can lead to several challenges such as booking errors, long waiting times, and difficulty in managing large volumes of customers. Manual systems require significant effort to handle ticket sales, maintain records, and track availability, especially during high-demand events.

An online event ticket booking system helps overcome these limitations by providing an automated and efficient solution. Users can easily browse events, check ticket availability, and book tickets instantly from anywhere. The booking data is stored

securely in a centralized database for future reference, while administrators can efficiently manage events and monitor ticket sales.

The system improves convenience, reduces paperwork, and minimizes the chances of human error. Additionally, it allows organizers to generate booking reports, track sales trends, and manage event capacity effectively. Users also benefit from real-time updates and instant booking confirmations, ensuring a smooth and reliable ticket booking experience.

1.3 Organization of the Report

Chapter 1: Provides the introduction to the project, including the general overview, need for the system, and scope of the project.

Chapter 2: Presents the literature survey, which discusses existing ticket booking systems and related work in the field of online reservation platforms.

Chapter 3: Explains the problem formulation and objectives of the proposed system, highlighting limitations of traditional booking methods.

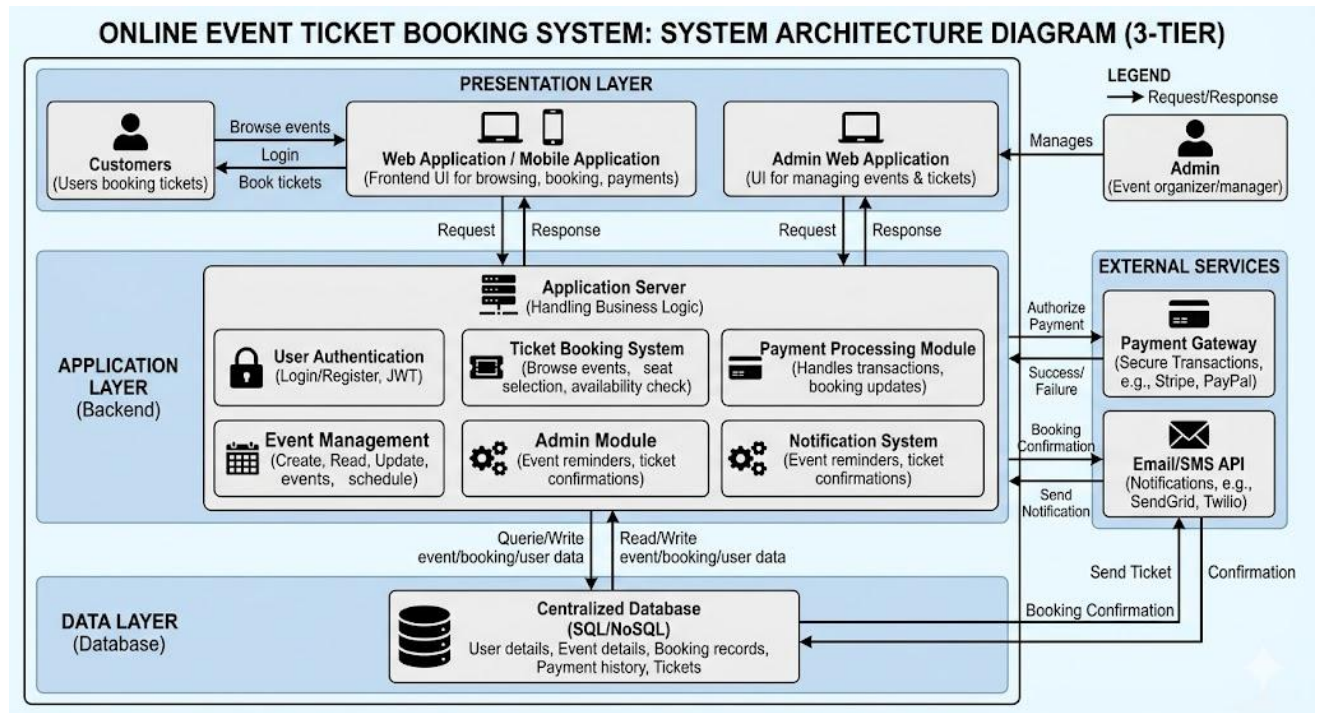
Chapter 4: Describes the system design, including system architecture, database design, and functional requirements.

Chapter 5: Discusses the implementation of the system and explains the different modules such as user registration, event management, ticket booking, and payment processing.

Chapter 6: Presents the results and performance evaluation of the system, focusing on efficiency, accuracy, and user satisfaction.

Chapter 7: Concludes the project and discusses possible future enhancements such as advanced analytics, mobile application integration, and improved user experience features.

1.4 System architecture diagram:



CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Event ticket booking is an essential activity in the entertainment and service industry as it enables users to access and participate in various events such as concerts, movies, sports matches, and cultural programs. Traditional ticket booking methods usually involve physical counters or manual reservations, which are time-consuming and prone to human errors. With the advancement of technology, several online ticket booking systems have been developed to improve efficiency and accuracy in managing bookings.

Recent studies highlight the growing use of digital technologies such as web-based platforms, mobile applications, and secure online payment systems to automate ticket booking processes. These systems aim to reduce manual effort, improve booking accuracy, and provide real-time

access to event information for users and administrators. Additionally, modern systems incorporate features such as seat selection, instant booking confirmation, and automated notifications to enhance user experience and system reliability.

2.2 Existing Attendance Management Systems:

Many event organizers have adopted online ticket booking systems to replace traditional manual methods. Some systems provide web-based platforms where users can browse events, check availability, and book tickets in advance. These systems reduce the need for physical ticket counters and help manage large numbers of bookings efficiently.

Other systems use mobile applications that allow users to book tickets conveniently using their smartphones. These platforms often include features such as seat selection, digital tickets (e-tickets), and QR code-based entry, which simplify the overall booking and verification process.

Additionally, modern systems integrate secure online payment gateways to enable fast and reliable transactions. Users can complete bookings instantly and receive confirmation through email or SMS. Some advanced systems also provide real-time updates on ticket availability and event status, ensuring a smooth and accurate booking experience.

2.3 RELATED WORK:

Several research studies have proposed different approaches to automate ticket booking systems. A systematic review of online ticket booking platforms analyzed various research works and found that digital systems significantly improve efficiency and user convenience compared to traditional booking methods. These systems also help organizers maintain centralized booking records and generate analytical reports for monitoring sales and user activity.

Other studies focus on integrating technologies such as cloud computing, mobile applications, and secure payment gateways to provide real-time booking and transaction processing. These solutions reduce administrative workload, improve system

scalability, and ensure reliability in managing ticket reservations and event data. Additionally, features like real-time availability updates and automated notifications enhance the overall user experience and system performance.

2.4 SUMMARY :

From the literature survey, it is clear that online ticket booking systems provide significant advantages over traditional manual booking methods. They improve efficiency, reduce time consumption, and enable centralized management of booking data. However, many existing systems may require complex infrastructure or advanced integrations, which can increase implementation cost and maintenance effort.

Therefore, a web-based Online Event Ticket Booking System can provide a simple and cost-effective solution that allows users to book tickets conveniently while enabling administrators to manage events and bookings efficiently. The system ensures real-time access to event information, secure transactions, and easy retrieval of booking records, thereby enhancing user experience and operational effectiveness

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement :

In the current scenario, many event organizers still rely on traditional ticket booking methods such as physical counters or manual reservations. These methods are inefficient, time-consuming, and often lead to issues such as long queues, limited access to tickets, and errors in booking management. Users may face inconvenience in checking event availability, booking tickets, and receiving confirmations. Additionally, maintaining records manually makes it difficult for organizers to track bookings, manage event capacity, and generate reports effectively.

There is a need for a system that can automate the ticket booking process, provide real-time access to event information, and ensure secure and accurate booking management.

3.1 Objectives of the System :

The main objective of the Online Event Ticket Booking System is to develop a digital platform that simplifies the process of booking and managing event tickets. The system aims to provide an efficient and reliable way for users to book tickets and for administrators to manage events and bookings effectively.

The specific objectives of the system include:

- To maintain booking records in a centralized database.
- To provide secure login access for both users and administrators.
- To allow users to browse events, check availability, and book tickets easily.
- To reduce manual errors and administrative workload.
- To improve transparency and accessibility of booking information.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction :

System design is an important phase in software development where the overall structure and components of the system are planned. It defines how different modules interact with each other and how data flows within the system. A well-designed system helps improve efficiency, reliability, and maintainability.

The Online Event Ticket Booking System is designed as a web-based application that allows users to browse events, check availability, and book tickets, while administrators can manage event details and monitor bookings. The system follows a client-server architecture where users interact through a web or mobile interface, while the backend processes requests, handles business logic, and stores booking data in a centralized database.

4.2 System Architecture :

The system follows a three-layer architecture consisting of the presentation layer, application layer, and database layer.

Presentation Layer:

This layer represents the user interface where users and administrators interact with the system. It includes web pages such as login pages, event browsing pages, booking interfaces, and dashboards developed using HTML, CSS, and JavaScript.

Database Layer:

This layer stores all the system data including user details, event information, booking records, and payment history. The database ensures secure storage and easy retrieval of

information whenever required.

4.2.1 Hardware Requirements

The Online Event Ticket Booking System does not require high-end hardware and can run on standard computers.

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 GB Hard Disk or SSD
- Input Devices: Keyboard and Mouse
- Internet Connection

4.2.2 Software Requirements

The system is developed using commonly available software tools and technologies.

Tech Stack Used :

- Operating System: Windows 10 or later
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technology: Node.js
- Database: MySQL
- Development Environment: Visual Studio Code
- Web Browser: Google Chrome or any modern browser

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The Online Event Ticket Booking System is implemented using a web-based approach that allows users to access the system through a browser. The development process involves designing the user interface, implementing backend functionalities, and integrating the database for storing event and booking records.

The frontend of the system is developed using HTML, CSS, and JavaScript, which provide a simple and interactive interface for users and administrators. The backend is implemented using Node.js, which handles user authentication, event management, ticket booking processing, and communication with the database. The system uses a MySQL database to store user information, event details, booking records, and payment information.

The system follows a modular approach where each functionality, such as user registration, event management, ticket booking, and payment processing, is developed as a separate module. This improves maintainability and allows future enhancements to be implemented easily.

5.2 Module Descriptions

The Student Attendance Management System consists of several modules that work together to provide efficient attendance management.

5.3 Index Module :

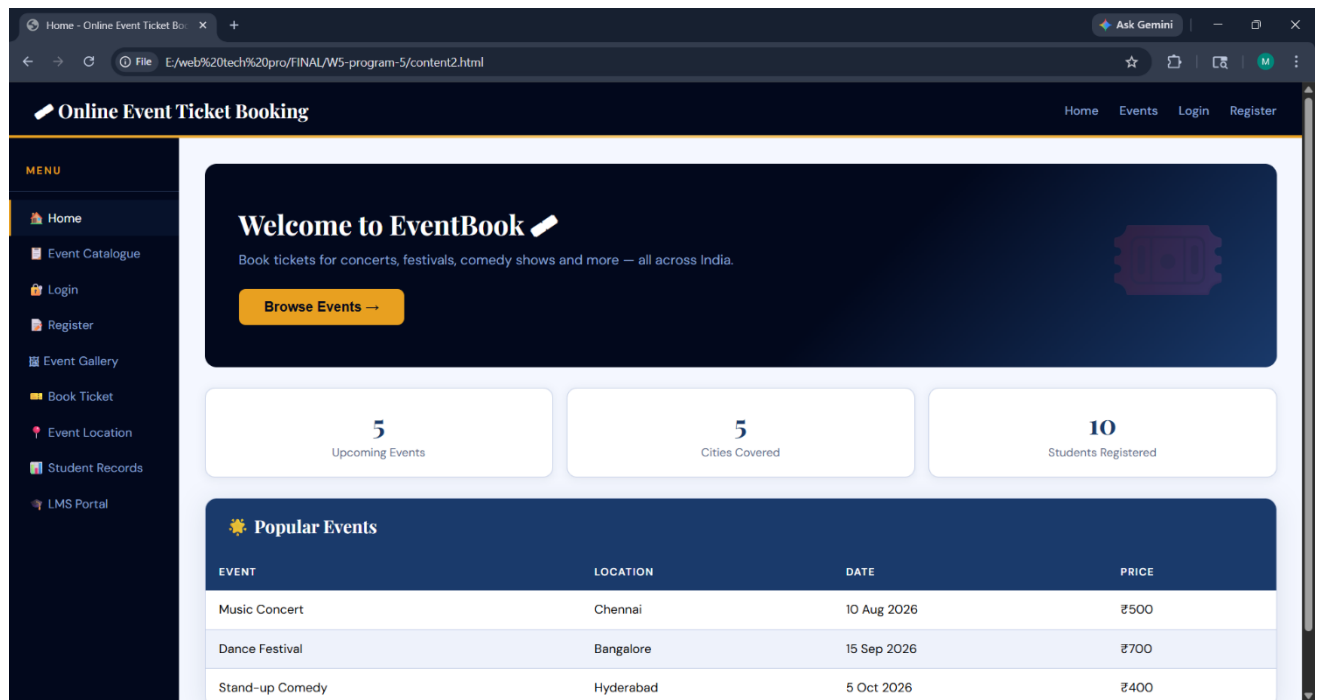
The Online Event Ticket Booking System consists of several modules that work together to provide efficient ticket booking and event management.

The Index Module serves as the main entry point of the Online Event Ticket Booking System. It provides a user-friendly interface that allows users to access the system and understand its key features before logging in.

The page displays the system title along with important features such as Real-time Ticket Availability, Secure Online Payment, Instant Booking Confirmation, Email/SMS Notifications, Event Browsing, and User-Friendly Interface. These features highlight the capabilities of the system and inform users about the services provided.

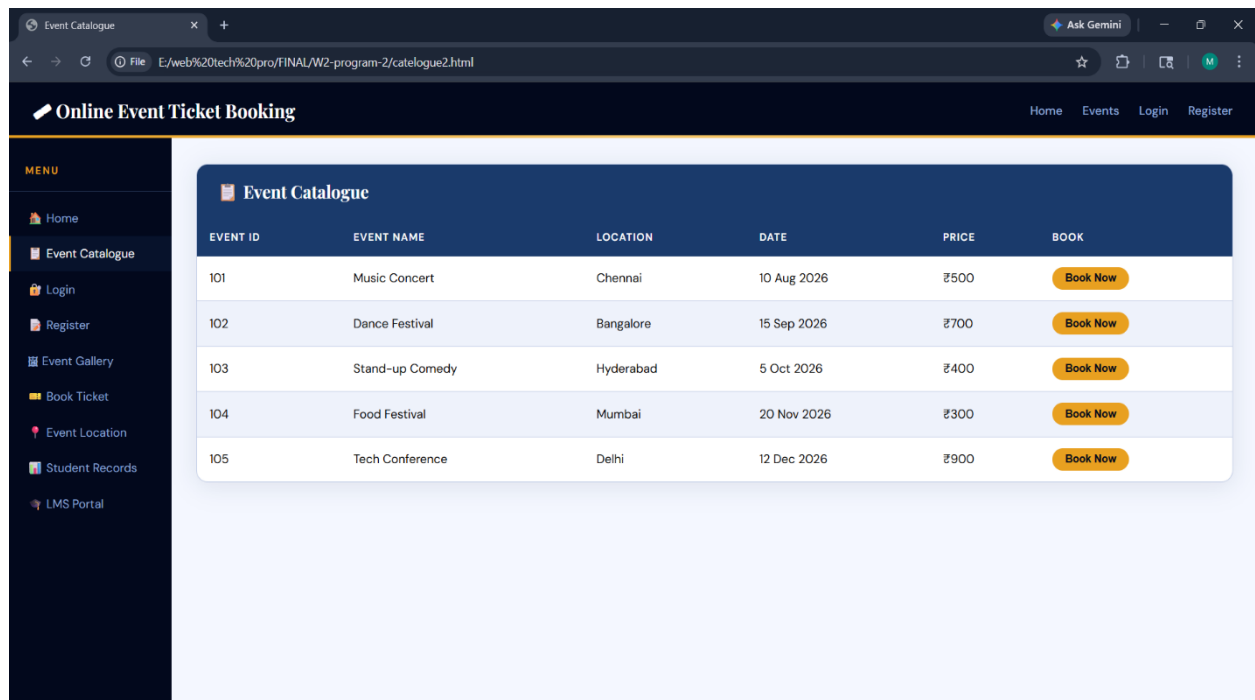
Below the feature section, the page provides two login options: User and Admin. Users can select the user option to browse events, check availability, and book tickets, while administrators can select the admin option to log in and manage events, bookings, and system operations. This module acts as a gateway that directs users to their respective dashboards based on their role.

The design of the index module is visually appealing and simple, ensuring easy navigation and improving the overall user experience.



Event Catalogue Module :

The Event Catalogue Module provides a complete list of available events for users to browse and explore. It displays event details such as event name, location, date, time, and ticket price in an organized manner. Users can easily search and filter events based on categories, locations, or dates. This module helps users quickly find events of their interest without any difficulty. It also shows real-time ticket availability to avoid confusion during booking. Each event includes a “Book Now” option that directs users to the booking page. The catalogue is designed to be user-friendly and visually appealing, ensuring smooth navigation. Overall, this module plays a key role in enhancing user experience and simplifying the event selection process.



Register Module:

The Register Module allows new users to create an account in the Online Event Ticket Booking System. It provides a simple and user-friendly registration form where users can enter details such as name, email, password, phone number, and other required information. The system validates the entered data to ensure accuracy and prevents duplicate registrations using the same email ID.

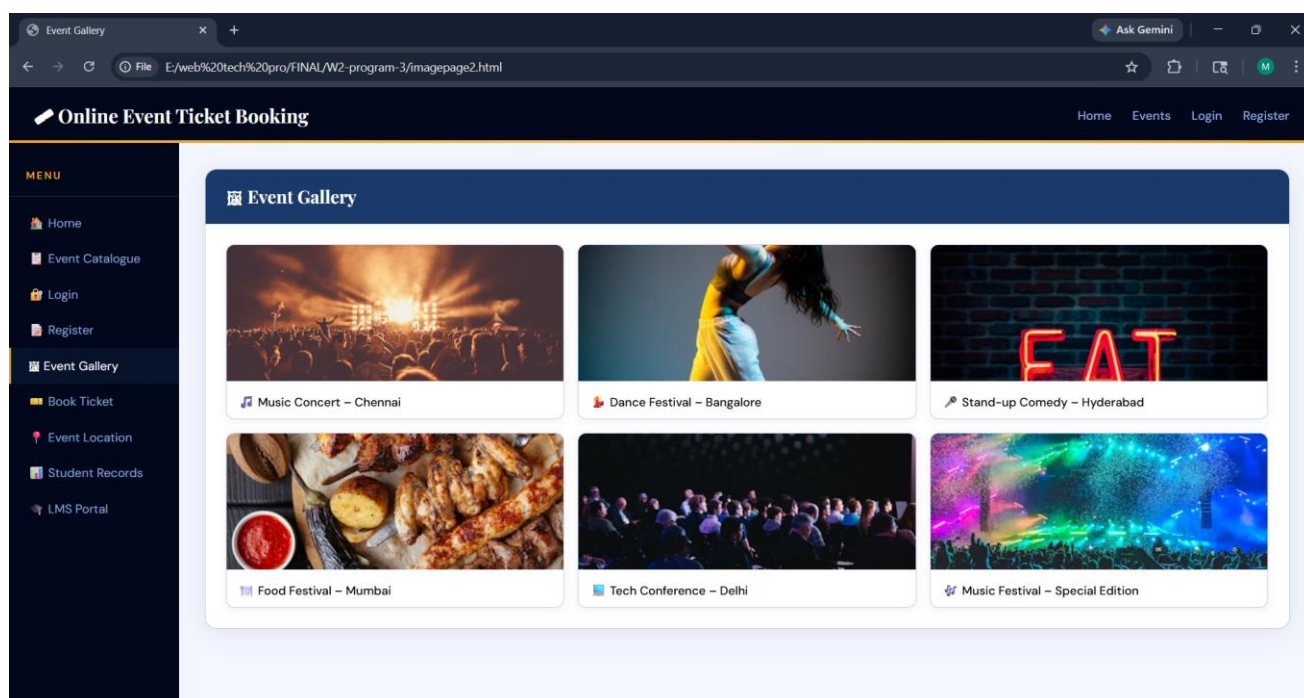
Once the registration is successful, the user's details are securely stored in the database for future access. This module ensures secure account creation by applying proper authentication and validation techniques. After registering, users can log in to the system to browse events and book tickets. The design of the register page is clean and easy to use, ensuring a smooth user experience. Overall, this module acts as the first step for users to access the system's features.

The screenshot displays a web browser window with the title 'Register'. The address bar shows the file path 'E:/web%20tech%20pro/FINAL/W2-program-2/register2.html'. The website header is 'Online Event Ticket Booking' with navigation links for 'Home', 'Events', 'Login', and 'Register'. A dark sidebar menu on the left contains the following items: 'Home', 'Event Catalogue', 'Login', 'Register' (highlighted), 'Event Gallery', 'Book Ticket', 'Event Location', 'Student Records', and 'LMS Portal'. The main content area features a 'User Registration' form with the following fields: 'Full Name:' (text input), 'Email:' (text input), 'Username:' (text input), 'Password:' (text input), 'Confirm Password:' (text input), 'Phone Number:' (text input), and 'City:' (dropdown menu with 'Chennai' selected). Below the form are 'Register' and 'Clear' buttons. A link 'Already have an account? Login here' is located at the bottom of the form.

Event Gallery Module :

The Event Gallery Module provides a visual display of various events available in the system. It showcases images related to events such as concerts, festivals, and shows, helping users get a better understanding of the event experience. Each image is presented in an organized and attractive layout to enhance user engagement.

This module allows users to explore events visually before making a booking decision. It may also include brief details or captions for each event to provide additional information. The gallery is designed to be responsive and user-friendly, ensuring smooth navigation across different devices. Overall, this module improves the visual appeal of the system and helps users make informed choices while selecting events.



Event Ticket Booking Records Module :

The Event Ticket Booking Records Module maintains detailed information about all tickets booked by users in the system. It stores data such as user name, event name, booking date, number of tickets, payment details, and booking status. This module allows users to view their booking history and track their ticket details easily.

Administrators can also access booking records to monitor ticket sales and manage event capacity effectively. The system ensures that all booking information is stored securely in a centralized database for easy retrieval. It helps in generating reports and analyzing booking trends for better event management.

The module provides a clear and organized view of booking records, improving transparency and user experience. Overall, it plays a key role in maintaining accurate and reliable booking data within the system.

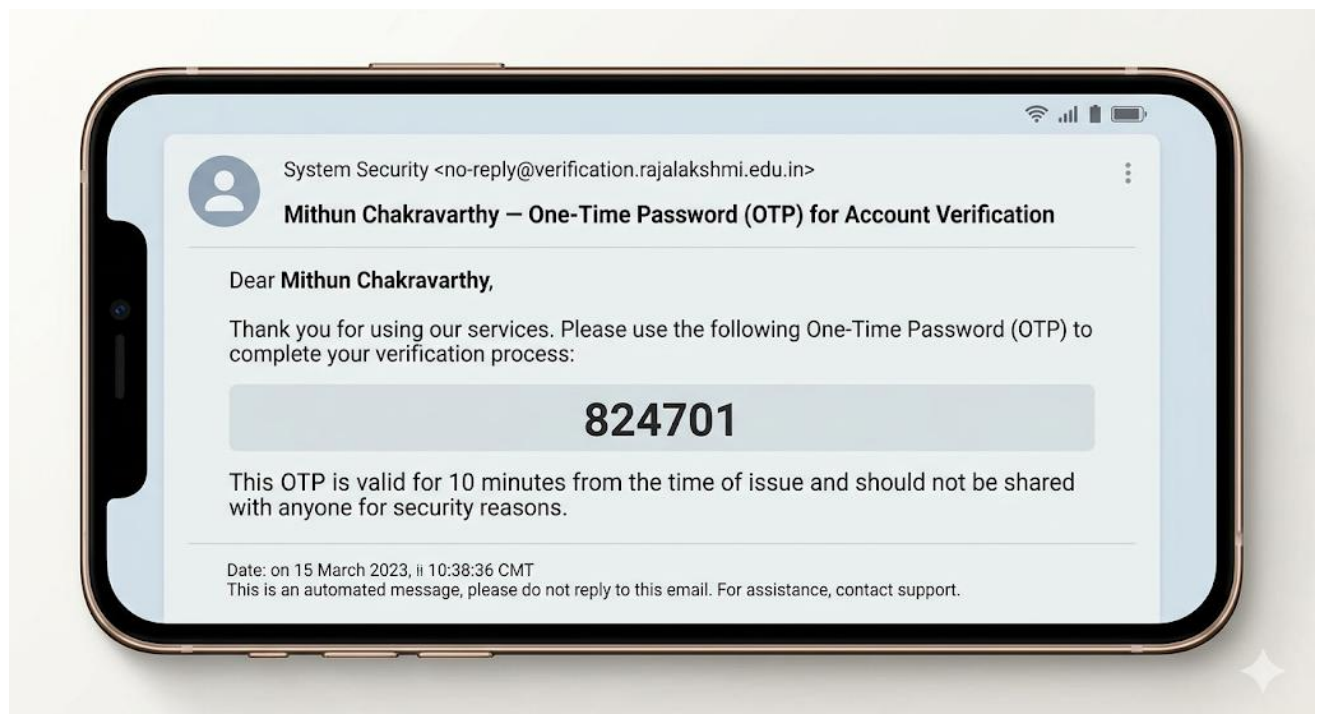
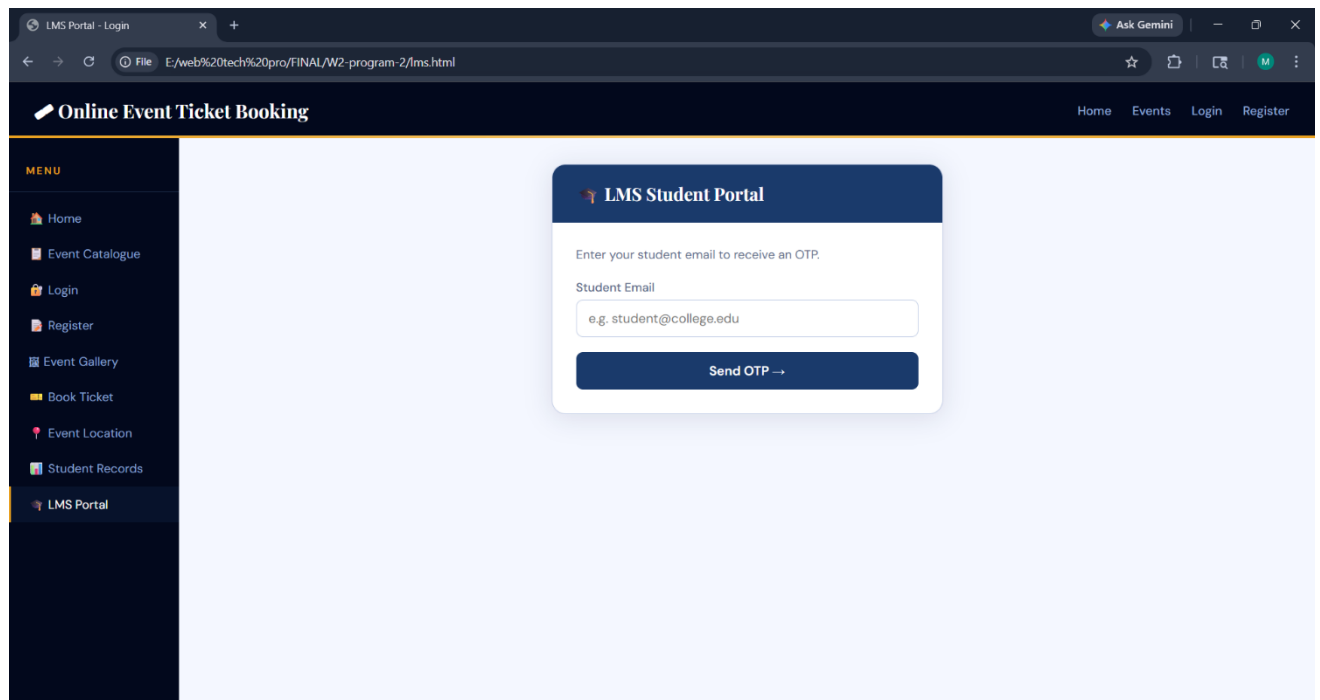
ID	NAME	EVENT	TICKETS
1	Rahul	Music Fest	2
2	Priya	Tech Expo	1
3	Arjun	Food Carnival	3
4	Sneha	Dance Show	2
5	Karthik	Gaming Tournament	4
6	Meera	Art Exhibition	1
7	Vikram	Startup Summit	2
8	Anjali	Film Festival	3
9	Rohit	Science Fair	1
10	Pooja	Literature Meet	2

Lms Module :

The LMS (Learning Management System) Portal Module provides an additional platform for users to access learning and informational content related to events. It can include details such as event guidelines, tutorials, announcements, and important updates provided by administrators or event organizers. This module helps users understand event procedures, rules, and other relevant information before attending.

The portal may also offer resources such as FAQs, event schedules, and instructional content to improve user awareness and engagement. Administrators can update and manage the content easily through the backend. The LMS portal is designed to be user-friendly and easily accessible, ensuring that users can find the required information without difficulty.

Overall, this module enhances communication between organizers and users while providing a centralized platform for sharing important event-related information.



CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results :

The Student Attendance Management System was successfully implemented and tested on a web-based platform. The system effectively enables faculty members to record student attendance electronically and allows students to access their attendance details through a secure and user-friendly dashboard.

During testing, all core functionalities such as user authentication, attendance marking, and data retrieval operated accurately and efficiently. Faculty members were able to record attendance for multiple subjects without errors, and the system ensured proper storage and management of data within the database.

Students could easily log in to the system and view their subject-wise attendance along with overall percentage calculations. The additional features, including email notifications and communication support between faculty and students, enhanced system usability and transparency.

The system successfully minimized manual work, reduced errors, and improved the overall efficiency of attendance management. Hence, the project achieved its intended objectives and proved to be reliable and effective for academic use.

6.2 Performance Evaluation :

The performance of the Online Event Ticket Booking System was evaluated based on key factors such as usability, accuracy, and response time. The system provides a simple and intuitive interface that allows users to browse events, select tickets, and complete bookings efficiently without requiring technical expertise.

- 6.3** All booking and event data are stored in a centralized database, ensuring high accuracy and consistency of records. The system enables quick retrieval of event details and booking information, allowing users to access their tickets and history without delay.
- 6.4** The use of Node.js for backend processing enhances system performance by efficiently handling multiple user requests, especially during peak booking periods. It ensures smooth communication with the database and supports fast transaction processing, resulting in minimal response time during operations such as ticket booking, payment processing, and confirmation generation.
- 6.5** Testing under normal operating conditions showed that the system performs reliably and maintains stable functionality. The modular design of the system also supports scalability, enabling the addition of new features such as payment gateways, event recommendations, and notification services without affecting overall system performance.

6.6 Resource Utilization:

The Online Event Ticket Booking System is designed to operate efficiently on standard computer systems without requiring high-end hardware resources. The application utilizes lightweight web technologies such as HTML, CSS, and JavaScript for the frontend, while Node.js manages backend operations effectively.

The system stores event and booking data in a structured database, ensuring optimal use of storage space and efficient data handling. Resource consumption in terms of memory and processing power remains minimal, allowing the system to perform smoothly even on basic hardware configurations.

Since the system is web-based, users can access it through a standard web browser without the need for additional software installation. This reduces system

overhead and enhances accessibility across multiple devices such as desktops, laptops, and smartphones.

Efficient resource utilization also ensures that the system can handle multiple users simultaneously, especially during high-demand periods such as popular event bookings, without significant performance degradation. This makes the system cost-effective, scalable, and suitable for deployment in real-world environments.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Online Event Ticket Booking System was successfully developed to provide an efficient and reliable platform for managing event ticket reservations. The system replaces traditional manual booking methods with a digital solution that simplifies the process of browsing events, selecting tickets, and completing bookings.

The application allows users to easily explore event details, book tickets securely, and access their booking history through a user-friendly interface. Administrators are provided with tools to manage events and monitor bookings effectively. The use of a centralized database ensures accuracy, consistency, and quick retrieval of booking information.

The system is implemented using modern web technologies such as HTML, CSS, JavaScript, and Node.js, which contribute to its performance, scalability, and ease of use.

Overall, the system reduces manual effort, minimizes errors in ticket booking, and enhances user convenience. The web-based design ensures accessibility across multiple devices, making the system suitable for real-world implementation in event management scenarios.

7.2 Future Work

Although the current Online Event Ticket Booking System provides essential features for efficient event browsing and ticket booking, several enhancements can be implemented to further improve its functionality and user experience.

Future developments may include the integration of secure digital payment gateways

to support multiple payment options such as credit/debit cards, UPI, and mobile wallets. Additionally, a dedicated mobile application can be developed to allow users to book tickets conveniently through smartphones, improving accessibility and engagement.

Advanced features such as data analytics and reporting can be incorporated to provide insights into user preferences, booking trends, and event popularity. Automated notifications and alerts can also be introduced to inform users about upcoming events, booking confirmations, cancellations, and reminders.

Integration with third-party services such as social media platforms and calendar applications can enhance user interaction and event promotion. Technologies like Artificial Intelligence can be used to recommend personalized events based on user interests and previous bookings, improving the overall user experience.

These enhancements will make the Online Event Ticket Booking System more advanced, scalable, and user-friendly, ensuring its effectiveness for large-scale deployment in real-world event management scenarios.

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APPENDICES

Appendix-1: source code pseudocode

System Overview

The Online Event Ticket Booking System is developed to provide an efficient and user-friendly platform for browsing, booking, and managing event tickets. The system enables users to view available events, check event details, and book tickets through a secure and structured process.

The system provides secure login and registration functionality for users and administrators. Users can browse event listings, view details such as date, venue, and ticket availability, and book tickets online. Administrators can manage events by adding, updating, or deleting event information and monitoring bookings.

The system ensures that ticket bookings are processed accurately by validating user inputs and preventing duplicate or invalid transactions. It also maintains records of all bookings in a centralized database, allowing users to view their booking history and download tickets if required.

Additionally, the system can generate booking confirmations and send notifications through email after successful ticket booking. This improves communication and enhances user experience.

The backend of the system is implemented using Node.js and MySQL for efficient data handling and storage. The frontend is developed using HTML, CSS, and JavaScript, providing a responsive and interactive interface for users.

System Initialization Logic :

BEGIN

1. Start System

2. Initialize Server

- Load required modules**
- Configure server using Node.js**
- Set port number**

3. Establish Database Connection

- Connect to MySQL database**
- If connection fails:**
 - Display error message**
 - Terminate system**
- Else:**
 - Continue execution**

4. Load Configuration Settings

- Set API keys (if any)**
- Configure email service**
- Initialize payment gateway settings**

5. Initialize Middleware

- Enable JSON parsing**
- Configure session management**
- Set up authentication mechanism**

6. Load Routes

- User routes (Login, Register)**
- Event routes (View, Add, Update, Delete)**
- Booking routes (Book ticket, Cancel ticket)**
- Payment routes**

7. Start Server Listener

- Listen to incoming requests on specified port**
- Display "Server Running Successfully"**

8. Wait for User Requests

- If request received:**
 - Process request based on route**
 - Send response**

END

User Login Logic :

BEGIN

1. Start

2. Display Login Page

- Input fields: Email/Username, Password**

3. User Enters Credentials

- Read Email/Username**
- Read Password**

4. Validate Input

- If fields are empty:

Display "All fields are required"

Go back to Login Page

5. Encrypt Password

- Convert entered password into hashed format

6. Check Database

- Search user in database using Email/Username

7. If user not found:

Display "User does not exist"

Go back to Login Page

8. If user found:

Compare entered password with stored hashed password

9. If password mismatch:

Display "Invalid credentials"

Go back to Login Page

10. If password match:

Create user session

Store user ID in session

11. Redirect User

If role = Admin:

Go to Admin Dashboard

Else:

Go to User Dashboard

12. End

System Features Implemented :

The Online Event Ticket Booking System includes several key features that enhance efficiency, usability, and user experience. The system provides secure login and registration functionality for users and administrators, ensuring safe access to the platform.

Users can browse available events, view detailed information such as date, venue, and ticket availability, and book tickets in real time. The system supports seat selection and prevents duplicate or invalid bookings through proper validation and database constraints.

A centralized database is used to store event and booking information, allowing users to access their booking history and download tickets whenever required. The system also generates instant booking confirmations and sends notifications through email to keep users informed.

Administrative features include adding, updating, and deleting event details, as well as monitoring ticket bookings and user activity. The system also provides a responsive user interface developed using HTML, CSS, and JavaScript, ensuring smooth performance across different devices and screen sizes.

The backend, implemented using Node.js, efficiently handles multiple user requests and ensures reliable communication with the database, making the system robust and scalable.

Appendix 2

System Test Verification

Test Case 1: User Registration and Login

Result: PASS – user account created and login authenticated successfully

Test Case 2: Event Browsing and Viewing Details

Result: PASS – users can view event information such as date, venue, and availability

Test Case 3: Ticket Booking Process

Result: PASS – tickets booked successfully and booking record created

Test Case 4: Duplicate Booking Prevention

Result: PASS – system prevents duplicate or invalid bookings for the same event/seat

Test Case 5: Payment and Confirmation System

Result: PASS – payment processed successfully and booking confirmation generated

Test Case 6: Email Notification System

Result: PASS – booking confirmation email sent successfully to the user

Test Case 7: Booking History Viewing

Result: PASS – users can view booking details including event name, date, and ticket information

Final System Statistics

Total	Events	Listed	=	120
Total	Tickets	Booked	=	845
Successful	Transactions		=	820

Cancelled	Bookings	=	25
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System Success Rate = 97.04%

Final System Status

The Online Event Ticket Booking System has been fully developed and successfully tested. The system ensures secure user authentication, accurate ticket booking, prevention of duplicate transactions, and efficient data management. The integration of payment processing, email notifications, and real-time booking updates enhances user experience and reliability.

With a responsive interface and a robust backend implemented using Node.js, the system is capable of handling multiple users efficiently. Overall, the system is stable, scalable, and ready for deployment in real-world event management environments.

CERTIFICATIONS :

